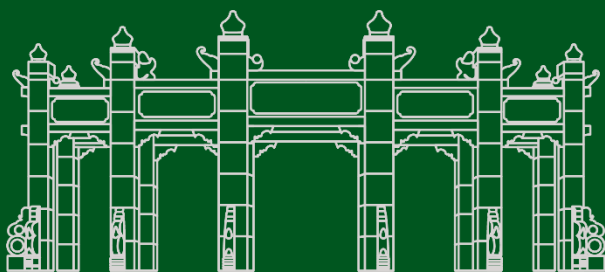


中山大學

Electron-Nucleus Cross Sections from Transfer Learning

冯仟潞



目录 Contents

01

Motivation

02

Electron-Carbon
Scattering by DNN

03

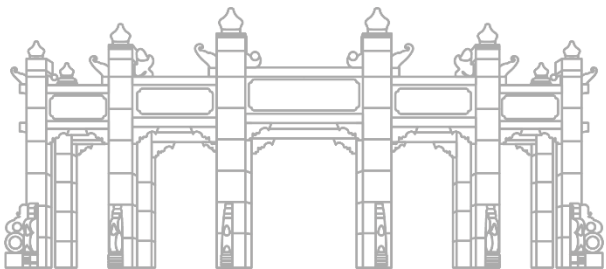
Electron-Nucleus
Scattering From TL

04

Conclusion

01

Motivation

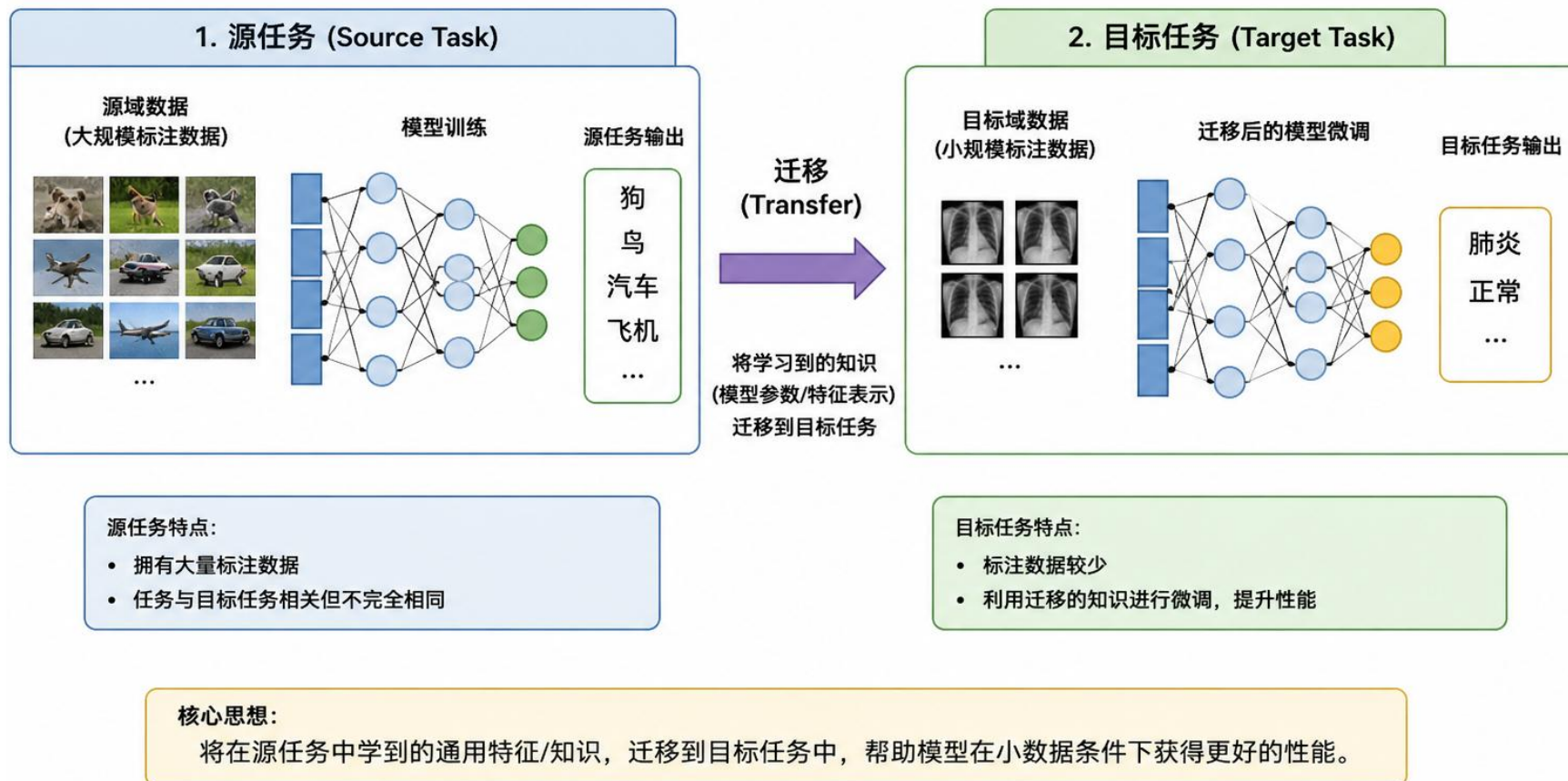


Motivation

- studies of neutrinos' fundamental properties
- complexity of the problem of neutrino interactions with nuclei in GeV energy region
 - different interaction mechanisms yield the same final states
 - the process of scattering involves multicomponent vector and axial currents
- similarities of electron and neutrino interactions at nuclear level
- scarcity of the electron-scattering measurements (eg: ^{16}O , ^{40}Ar)

Transfer learning allows a deep neural network (DNN) trained on one type of data to be adapted for new problems with limited information. We propose to use the transfer learning technique in physics. The DNN learns the details of one process, and after fine-tuning, it makes predictions for related processes. We consider the DNNs, trained on inclusive electron-carbon scattering data, and show that after fine-tuning, they accurately predict cross sections for electron interactions with nuclear targets ranging from helium-3 to iron.

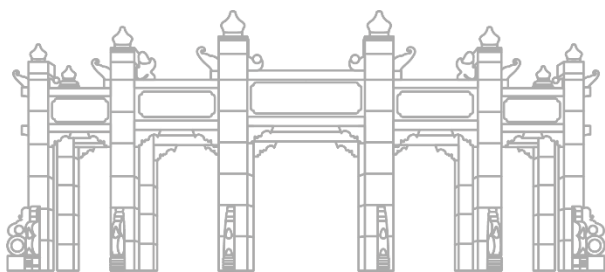
Transfer Learning



- source task: a large body of data for electron-scattering of carbon
- target task: the cross sections for nuclear targets ranging from helium-3 to iron

02

electron-carbon scattering by DNN

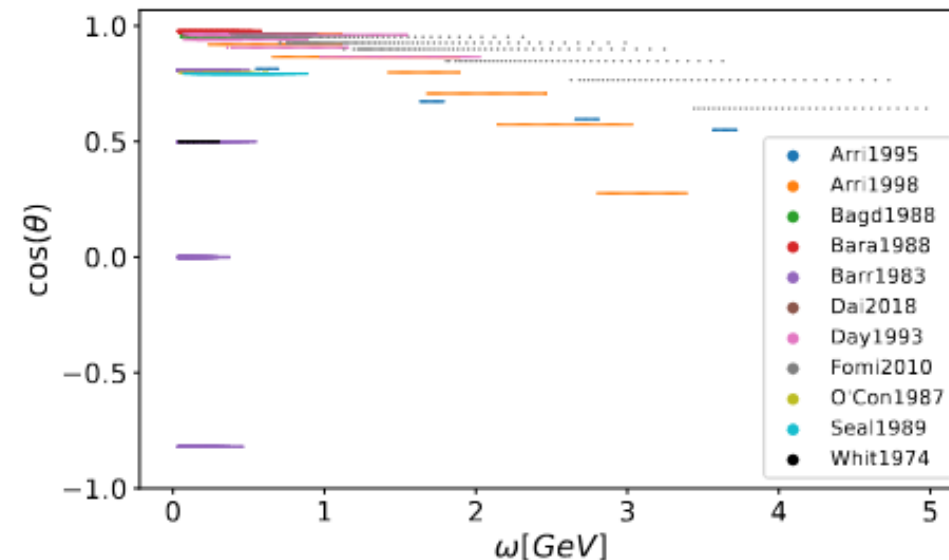
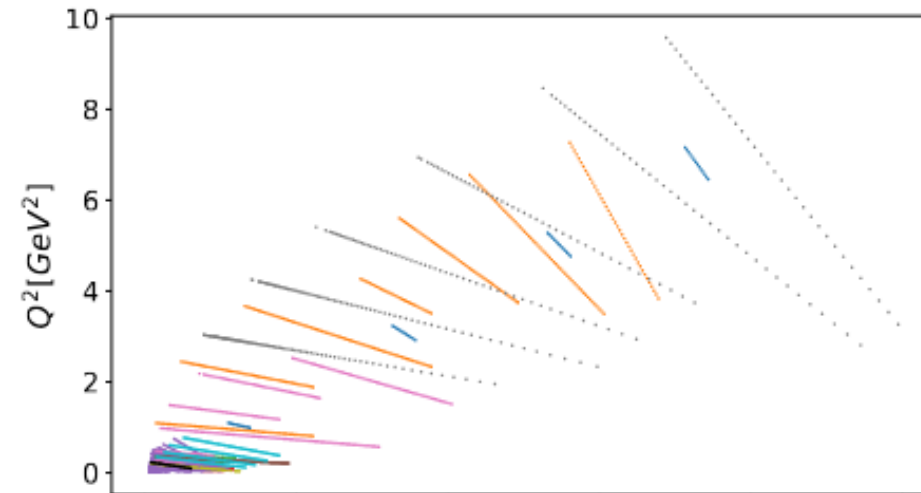


Electron-Scattering Data

- data: differential cross sections $d^2\sigma/d\omega d\Omega$
- kinematic region: from **quasielastic scattering** to **deep-inelastic scattering**

$$\omega > \omega_{\min} = \frac{M_A E_x + 0.5 E_x^2 + E^2 (1 - \cos \theta)}{M_A + E (1 - \cos \theta)}$$

Reference	Abbrev.	Norm. uncert.	Number of points
Arrington <i>et al.</i> [28]	Arri1995	4.0%	56
Arrington <i>et al.</i> [29]	Arri1998	4.0%	398
Bagdasaryan <i>et al.</i> [30]	Bagd1988	10.0%	125
Baran <i>et al.</i> [31]	Bara1988	3.7%	259
Barreau <i>et al.</i> [32]	Barr1983	2.0%	1243
Dai <i>et al.</i> [33]	Dai2018	2.2%	177
Day <i>et al.</i> [34]	Day1993	3.4%	316
Fomin <i>et al.</i> [35]	Fomi2010	4.0%	359
O'Connell <i>et al.</i> [36]	O'Con1987	5.0%	51
Sealock <i>et al.</i> [37]	Seal1989	2.5%	250
Whitney <i>et al.</i> [38]	Whit1974	3.0%	31
Total			3265



Likelihood Analysis

datasets

$$\mathcal{D}_k = \{(E_k^i, \theta_k^i, \omega_k^i, d\sigma_k^i, \Delta d\sigma_k^i) : i = 1, \dots, N_k\}$$

where $d\sigma_k^i$ and $\Delta d\sigma_k^i$ are the i th measurement in k th dataset and its corresponding uncertainty, N_k number of data points in k th dataset.

loss function

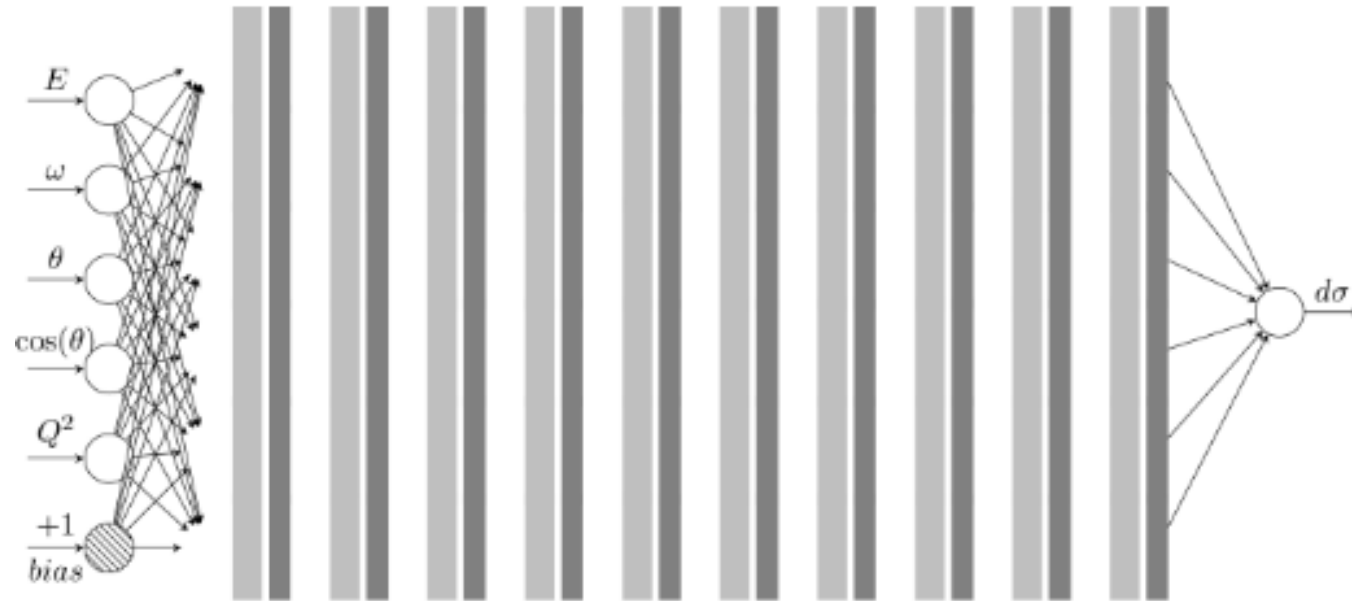
$$\chi_{\text{tot}} = \sum_{k=1}^{11} \left[\chi_k^2(\lambda_k) + \frac{1}{2} \left(\frac{1 - \lambda_k}{\Delta \lambda_k} \right)^2 \right]$$

$$\chi_k^2(\lambda_k) = \frac{1}{2} \sum_{i=1}^{N_k} \left(\frac{d\sigma_k^i - \lambda_k d\sigma_i^{\text{fit}}(E_k^i, \theta_k^i)}{\Delta d\sigma_k^i} \right)^2$$

$\Delta \lambda_k$ denoting the overall systematic uncertainty for k th dataset, and λ_k being a subject of optimization.



Neural Network Models



- **input layer:** 5-dimensional
- **10 blocks:** a hidden layer with 300 units fully connected with the previous module and batch-normalization layer, ReLU activation functions in each module
- **output layer:** given by a sigmoid function



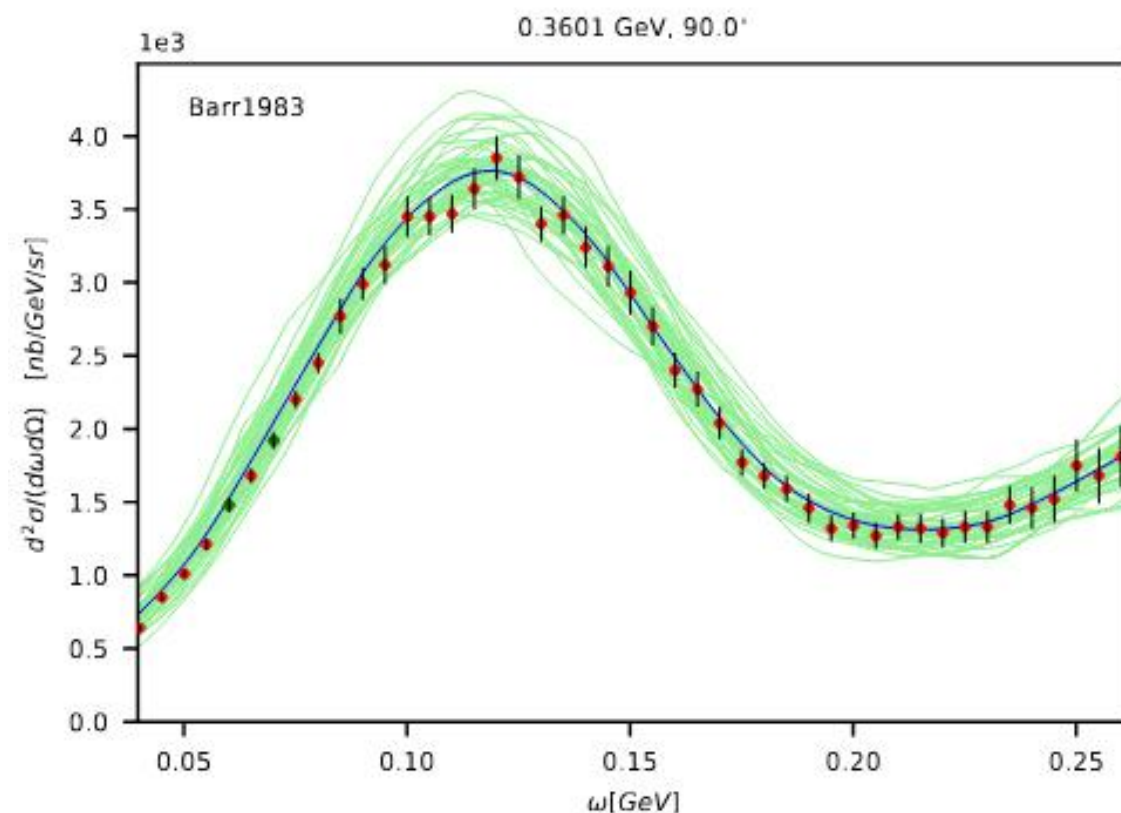
Uncertainty Estimation

bootstrap approach: drawn from the generating probability distribution

- sampling from a **normal distribution**

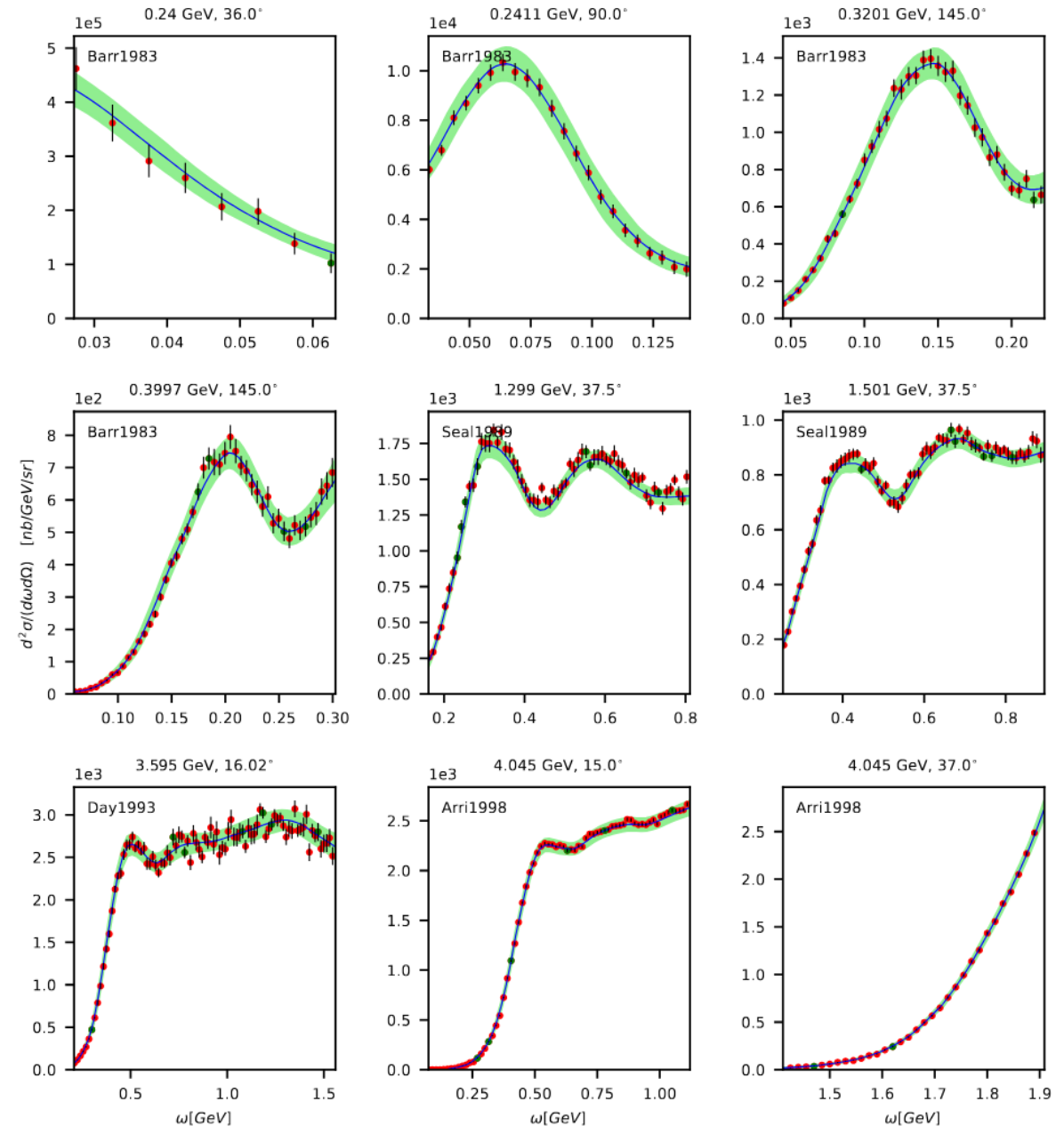
$$d\sigma_k^{\text{clone}, i} = d\sigma_k^i + r\Delta d\sigma_k^i$$

- split the full dataset into the training and test datasets in proportions 9 to 1
- 50 bootstrap datasets are created



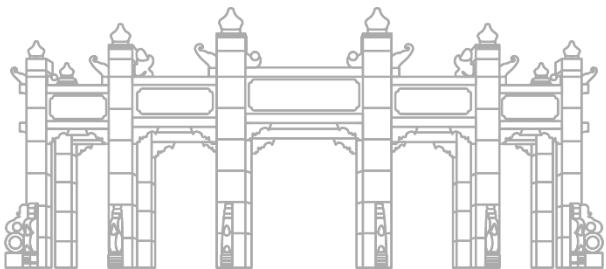
Numerical Result

- ran optimization for no more than about 60,000 epochs.
- used the AdamW algorithm to optimize
- keeping the value of weight decay as 0.004



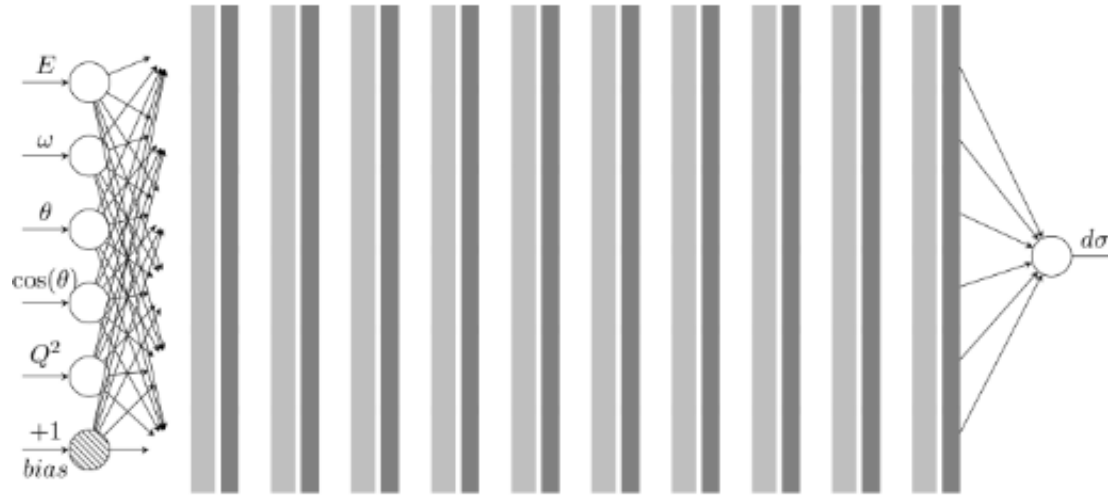
03

Electron-Nucleus Scattering From TL



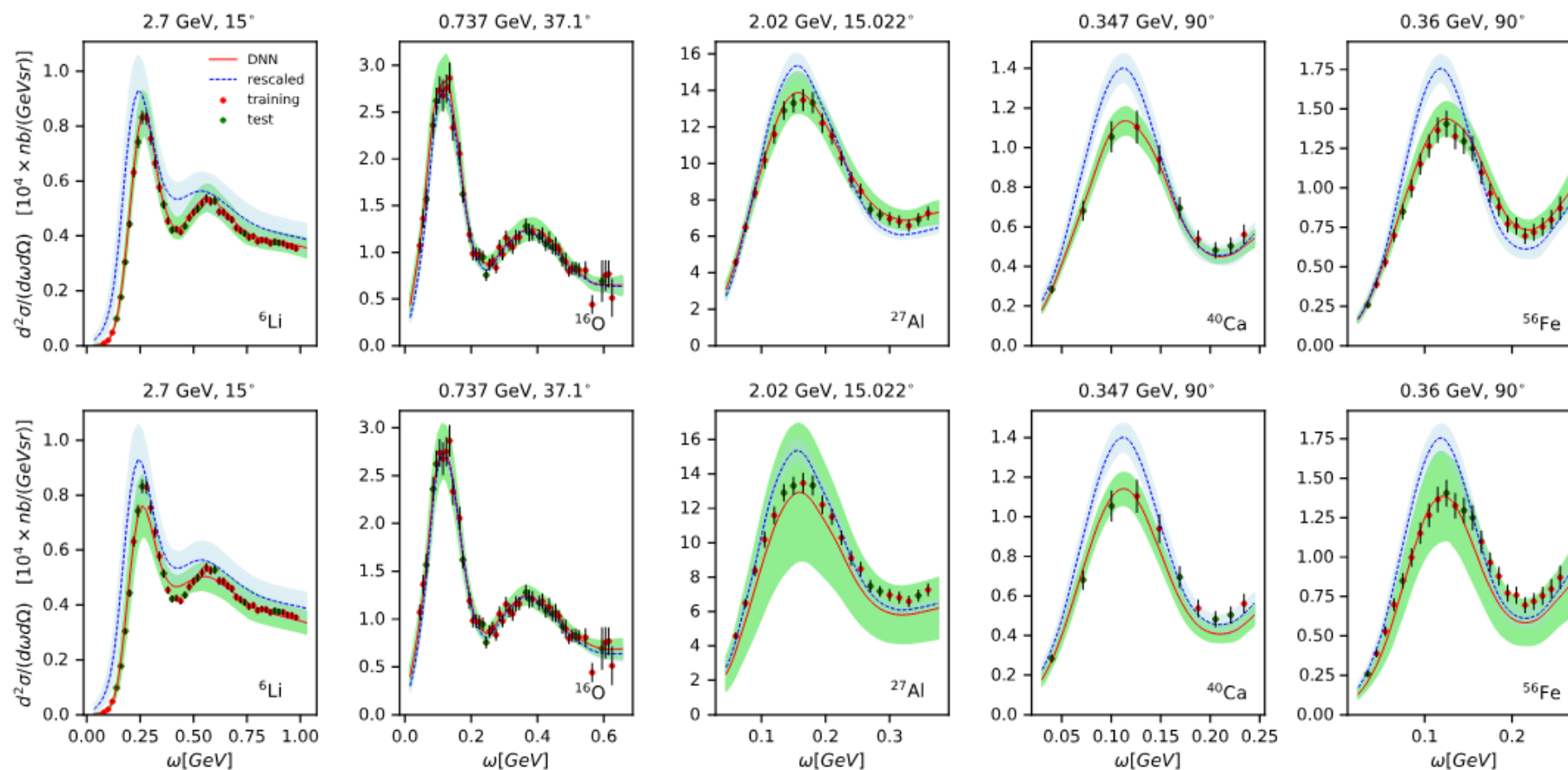
Framework

replace a pretrained network with a new one and training them with data from the new problem



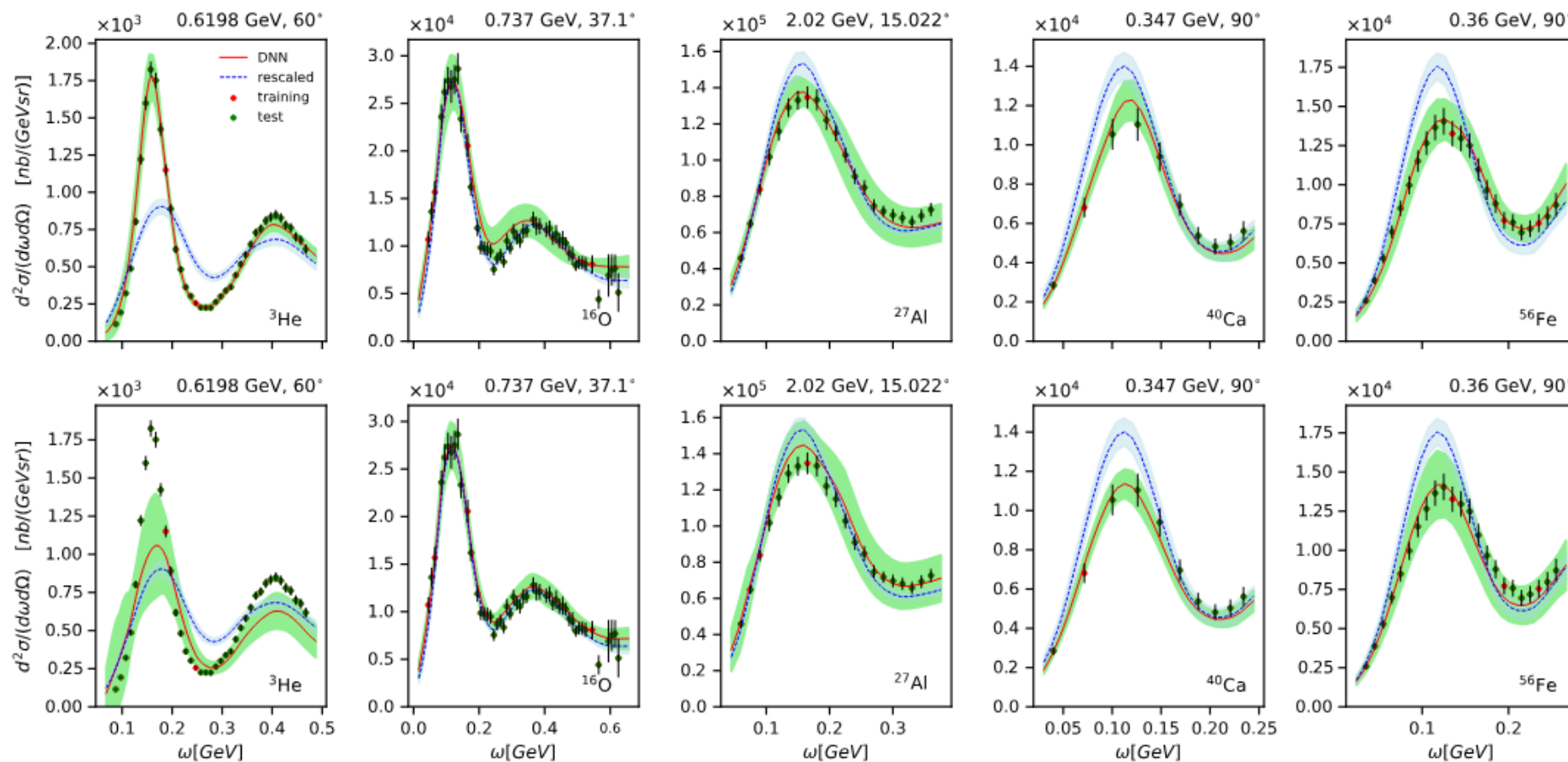
- the network, trained on electroncarbon data, has learned nuclear structure features that are broadly applicable to other medium-mass nuclei
- full fine-tuning: the parameters in all layers of the DNNs are optimized.
- shallow fine-tuning: the parameters in two last layers of the DNNs are optimized.
- rescaling: multiply the pretrained DNN output by the scale factor of $A/12$

the split of training and test set 7:3



top: full fine-tuning bottom: shallow fine-tuning

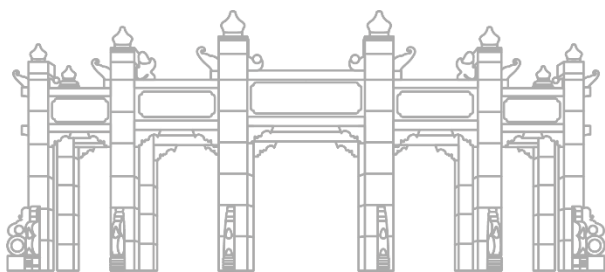
the split of training and test set 1:9



top: full fine-tuning bottom: shallow fine-tuning

04

Conclusion



Conclusion

- encode fundamental properties of the nuclear medium, which can be leveraged through the TL techniques to model related processes, even with limited information.
- significant structural differences between nuclei may reduce TL's efficiency

Further Application

neutrino-nucleus scattering

José L. Bonilla .etc Phys. Rev. D 112, 013007 (2025)

José L. Bonilla .etc Phys. Rev. D 113, 053001 (2026)



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谢谢观看

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